

# MAP & Conjugate Priors

MLE, with a prior's vote

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**Three heads in three flips**

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# Where Lecture 14 left us

L14 fixed the data, varied the parameter, and maximized the likelihood.

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} P(D \mid \theta)$$

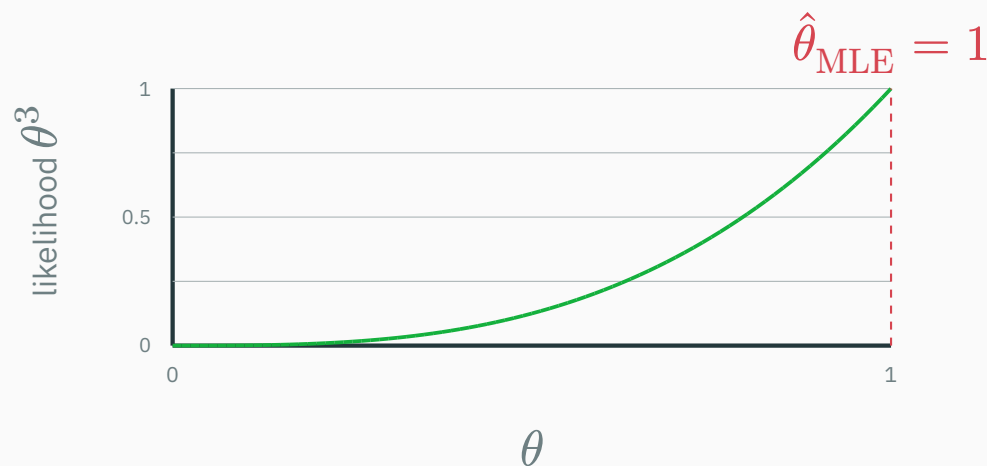
- The coin:  $n_H$  heads in  $n$  flips  $\Rightarrow \hat{\theta}_{\text{MLE}} = n_H/n$  — count and divide.
- With fixed-variance Gaussian observation noise, least squares and maximum likelihood have the same minimizer.
- Several common supervised losses are negative log-likelihoods for explicit observation models.

With only three observations, the Bernoulli MLE can lie on the boundary and report a zero probability for an unobserved outcome.

# Three flips, three heads

You flip a coin from your pocket three times. Heads, heads, heads. MLE does what it always does:

$$\hat{\theta}_{\text{MLE}} = \frac{n_H}{n} = \frac{3}{3} = 1.0$$



The plug-in model at  $\hat{\theta} = 1$  assigns probability zero to every future tail.

# Consequence of the boundary estimate

$\hat{\theta} = 1$  is a checkable claim: it prices “at least one tail in the next 100 flips” at **exactly zero**.

- Background knowledge about ordinary coins makes this plug-in prediction implausible.
- The three flips are real evidence — but they are **three flips**.
- A Bayesian model represents such pre-data information with a prior distribution.

**MLE has no slot for what you knew before the data. Today we build that slot: the **prior**.**

# Zero-count estimates in simple models

| System         | The three-heads bug, at work                                                                                                              |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| spam filter    | “invoice” never appeared in the spam folder →<br>$P(\text{invoice} \mid \text{spam}) = 0 \rightarrow$ one word acquits any email, forever |
| drug trial     | 3 successes in 3 patients → “this drug <i>cannot</i> fail”                                                                                |
| language model | a letter pair never seen in training → probability 0 → $\log 0 = -\infty$<br>loss on any text containing it                               |

L26 applies the corresponding Dirichlet update to character counts, so unseen transitions receive nonzero posterior-predictive probability.

# What would a fix even look like?

Three demands, straight from your intuition:

1. Tiny datasets must **not** produce absolute certainty.
2. For a fixed prior that assigns positive density near the data-generating parameter, its influence should diminish as evidence accumulates.
3. The hunch itself must be **adjustable** — a fairness zealot and a con-artist-spotter should be able to encode different starting beliefs.

Today's route: Bayes' rule defines the posterior; MAP extracts its mode; conjugacy gives closed-form posterior updates.

# Learning outcomes

By the end of this lecture you will be able to:

1. Read Bayes' rule as a **belief update**, naming all four factors and their jobs.
2. Compute **MAP** estimates and say exactly when  $\text{MAP} = \text{MLE}$ .
3. Use the **Beta distribution** as a prior over probabilities, reading  $\alpha, \beta$  as pseudo-counts.
4. Derive ★ the **Beta–Bernoulli conjugate update**,  $\text{Beta}(\alpha + n_H, \beta + n_T)$ .
5. Update **sequentially** as data streams in — and explain why order can't matter.
6. State the **Gaussian–Gaussian** posterior as a precision-weighted average.
7. Interpret **L2 regularization** as Gaussian-prior MAP (and L1 as Laplace-prior MAP).



# **Bayes' rule: the belief-update machine**

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# The four factors, named

One rule merges what you believed with what you saw:

$$p(\theta \mid D) = \frac{P(D \mid \theta) \cdot p(\theta)}{P(D)}$$

| Factor             | Name              | Its job                                                     |
|--------------------|-------------------|-------------------------------------------------------------|
| $p(\theta \mid D)$ | <b>posterior</b>  | density of $\theta$ <i>after</i> the data                   |
| $P(D \mid \theta)$ | <b>likelihood</b> | how loudly the data argues for each $\theta$ — L14's object |
| $p(\theta)$        | <b>prior</b>      | density of $\theta$ <i>before</i> the data                  |
| $P(D)$             | <b>evidence</b>   | one number that makes the left side a distribution          |

# Numbers first: a bag of coins

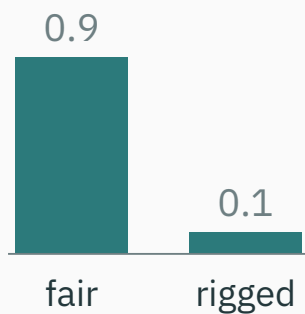
A bag holds 10 coins: **9 fair** ( $\theta = 0.5$ ), **1 rigged** ( $\theta = 0.9$ ). You draw one at random, flip it 3 times: **3 heads**. Did you draw the rigged one?

Two hypotheses  $\rightarrow$  Bayes' rule is a two-row table:

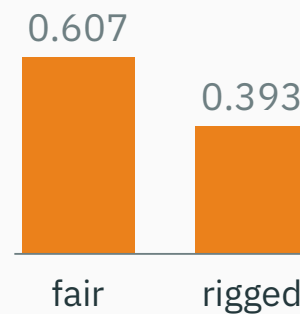
| Coin   | prior | likelihood $\theta^3$ | prior $\times$ lik | posterior                        |
|--------|-------|-----------------------|--------------------|----------------------------------|
| fair   | 0.9   | $0.5^3 = 0.125$       | 0.1125             | $0.1125/0.1854 = \mathbf{0.607}$ |
| rigged | 0.1   | $0.9^3 = 0.729$       | 0.0729             | $0.0729/0.1854 = \mathbf{0.393}$ |

Three heads moved the posterior probability of “rigged” from **10% to 39%**. The update multiplies prior by likelihood and rescales the two numbers to sum to 1.

before: **prior**



after 3 heads: **posterior**



```
prior = np.array([0.9, 0.1])      # P(fair), P(rigged)
like  = np.array([0.5**3, 0.9**3]) # P(HHH | each coin)
post  = prior * like              # multiply...
post /= post.sum()               # ...normalize -> [0.607, 0.393]
```

This finite-hypothesis example implements Bayes' rule by multiplication and normalization.

# The evidence is bookkeeping

For discrete hypotheses,  $P(D) = \sum_h P(D | h)P(h)$ . For a continuous parameter,  $P(D) = \int P(D | \theta)p(\theta) d\theta$ . In either case it is constant with respect to  $\theta$  and cannot move the posterior mode.

$$p(\theta | D) \propto P(D | \theta) \cdot p(\theta)$$

**Posterior  $\propto$  likelihood  $\times$  prior. The evidence is the normalization constant that makes the posterior integrate to one.**

The proportional form is sufficient whenever only the posterior shape or mode is needed.

# MAP: maximize the posterior density

Same recipe as MLE — but climb the **posterior** instead of the likelihood:

$$\theta_{\text{MLE}} = \arg \max_{\theta} P(D \mid \theta) \qquad \theta_{\text{MAP}} = \arg \max_{\theta} P(D \mid \theta) \cdot p(\theta)$$

In log-space (products  $\rightarrow$  sums — L2's rule, as always):

$$\theta_{\text{MAP}} = \arg \max_{\theta} \left[ \underbrace{\log P(D \mid \theta)}_{\text{log-likelihood}} + \underbrace{\log p(\theta)}_{\text{log-prior density}} \right]$$

**MAP = maximum a posteriori.** On the log scale, MAP maximizes the sum of log-likelihood and log-prior density.

# A flat prior changes nothing

Take  $p(\theta) = 1$  on  $[0, 1]$ . Then  $\log p(\theta) = 0$ :

$$\theta_{\text{MAP}} = \arg \max_{\theta} [\log P(D \mid \theta) + 0] = \theta_{\text{MLE}}$$

**A uniform prior on  $[0, 1]$  adds nothing to the log-posterior:  $\theta_{\text{MAP}} = \theta_{\text{MLE}}$ .**

This does not mean MLE assumes a prior: MLE and MAP are distinct procedures, and a density uniform in one parameterization need not stay uniform after reparameterization.

A prior changes the posterior and MAP estimate most visibly in small samples. For a fixed regular prior with support near the data-generating parameter, its influence on the MAP estimate typically diminishes as  $n$  grows.

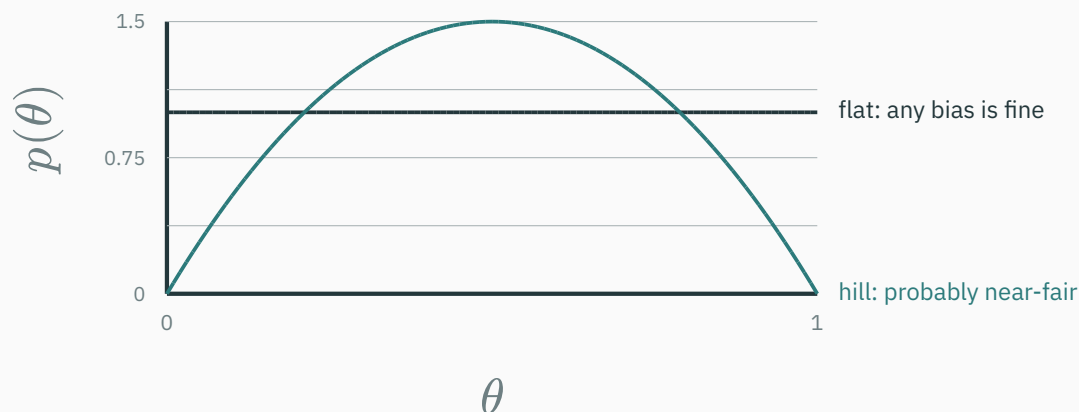
**The coin gets a prior**

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# A prior must live on $[0, 1]$

$\theta$  is a probability, so a prior is a **density on  $[0, 1]$**  (L12: continuous belief is area).  
Two candidates:



The hill is  $p(\theta) \propto \theta(1 - \theta)$  — zero at the impossible extremes, gentle peak at  $1/2$ . (It has a family name; we'll meet the family in a few slides.)

# Worked: the flat prior

Multiply the likelihood of three heads by the flat prior:

$$p(\theta \mid D) \propto \underbrace{\theta^3}_{\text{likelihood}} \cdot \underbrace{1}_{\text{prior}} = \theta^3$$

Climbing  $\theta^3$  on  $[0, 1]$  ends at the right wall:

$$\theta_{\text{MAP}} = 1.0 = \theta_{\text{MLE}}$$

The boundary estimate survives a flat prior. Moving it requires an informative prior.

## Worked: the hill prior, by hand

Same three heads, hill prior  $p(\theta) \propto \theta(1 - \theta)$ :

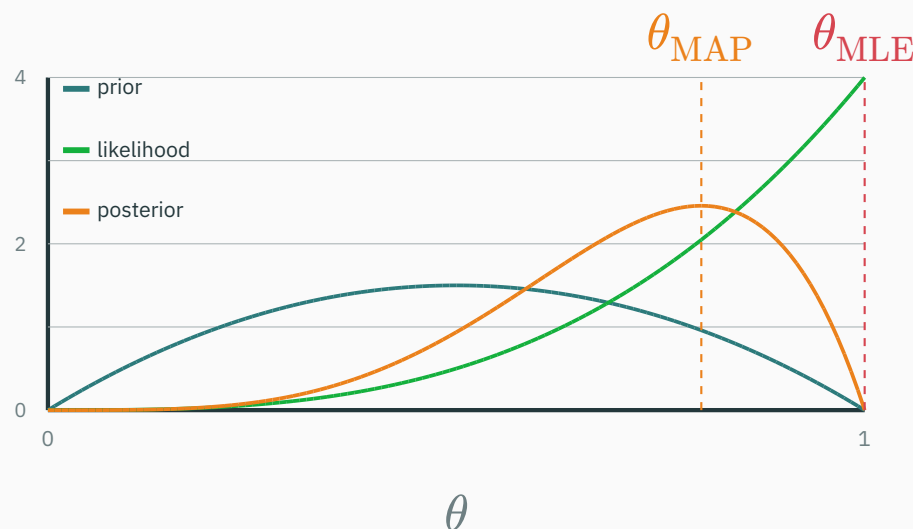
$$p(\theta \mid D) \propto \theta^3 \cdot \theta(1 - \theta) = \theta^4(1 - \theta)$$

Maximize with one derivative (drop constants; they can't move a peak):

$$\frac{d}{d\theta} [\theta^4(1 - \theta)] = 4\theta^3 - 5\theta^4 = \theta^3(4 - 5\theta) = 0$$

$$\Rightarrow \theta_{\text{MAP}} = \frac{4}{5} = 0.8$$

**Same three heads: the flat prior says 1.00; a gentle near-fair belief says 0.80.**



Likelihood rescaled to area 1 so the three shapes share one axis.

The posterior mode lies between the prior mode (0.5) and likelihood maximum (1.0), with the relative concentration of the two factors determining its position.

# Check it by computer

No calculus, just a grid — the same argmax, numerically:

```
th    = np.linspace(0, 1, 1001)
flat  = th**3                # likelihood x flat prior
hill  = th**3 * th * (1 - th) # likelihood x hill prior

th[flat.argmax()], th[hill.argmax()]    # -> (1.0, 0.8)
```

- Shapes only — no normalizing constants anywhere, because `argmax` ignores scale.
- This grid trick works for **any** prior you can evaluate — calculus optional. (T7 uses it to cross-check every derivation.)

A coin shows 9 heads in 10 flips. With a **flat** prior on  $\theta$ , the MAP estimate is...

**A.** 0.50 — the prior always pulls to fair

**B.** 0.90 — exactly the MLE

**C.** 0.83 — pulled part-way toward 1/2

**D.** 1.00 — MAP always sharpens the MLE

**Correct: B** —  $\theta_{\text{MAP}} = 0.90 = \theta_{\text{MLE}}$

**Why:** A flat prior adds  $\log 1 = 0$  to the log-posterior — the argmax cannot move, so  $\text{MAP} = \text{MLE} = 9/10$ . Answer C ( $\approx 0.83$ ) is what the *hill* prior  $\text{Beta}(2, 2)$  would give — right instinct, wrong prior; A forgets the data entirely; D isn't a thing MAP does.

# **The Beta family as a prior over probabilities**

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# The hill has a family name

Tilt the exponents and the hill becomes a whole family of beliefs on  $[0, 1]$ :

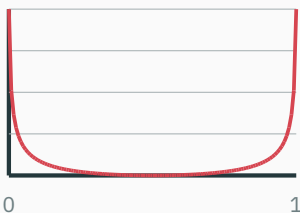
$$\text{Beta}(\theta; \alpha, \beta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha, \beta)}, \quad B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

- $\Gamma(k) = (k-1)!$  for positive integers, and the gamma function extends this definition so  $\alpha$  and  $\beta$  need not be integers.
- Our hill  $\propto \theta(1-\theta)$  is  $\text{Beta}(2, 2)$ ; the flat prior is  $\text{Beta}(1, 1)$ .

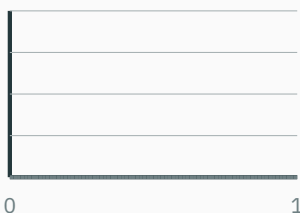
$B(\alpha, \beta)$  exists so the area is 1. We will **never** integrate it in this lecture — inspection will hand it to us for free.

All dials from  $\{0.5, 1, 2, 5\}$  — one family, six personalities:

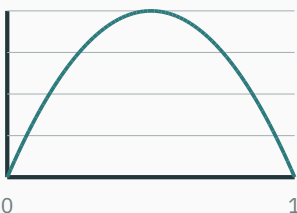
**Beta(0.5, 0.5) — extremist**



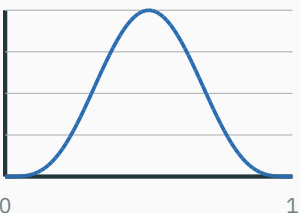
**Beta(1, 1) — agnostic**



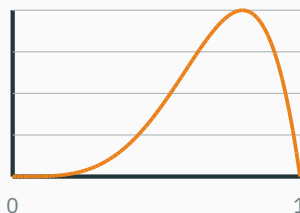
**Beta(2, 2) — gentle hunch**



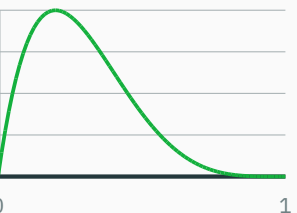
**Beta(5, 5) — firmer hunch**



**Beta(5, 2) — leans heads**



**Beta(2, 5) — leans tails**



One knob pair covers superstition, indifference, hunches, conviction, and lopsided suspicion.

# A mode-based pseudo-count mnemonic

For  $\alpha, \beta > 1$  the peak sits at  $\text{mode} = (\alpha - 1)/(\alpha + \beta - 2)$  — exactly the MLE of a record with  $\alpha - 1$  heads and  $\beta - 1$  tails.

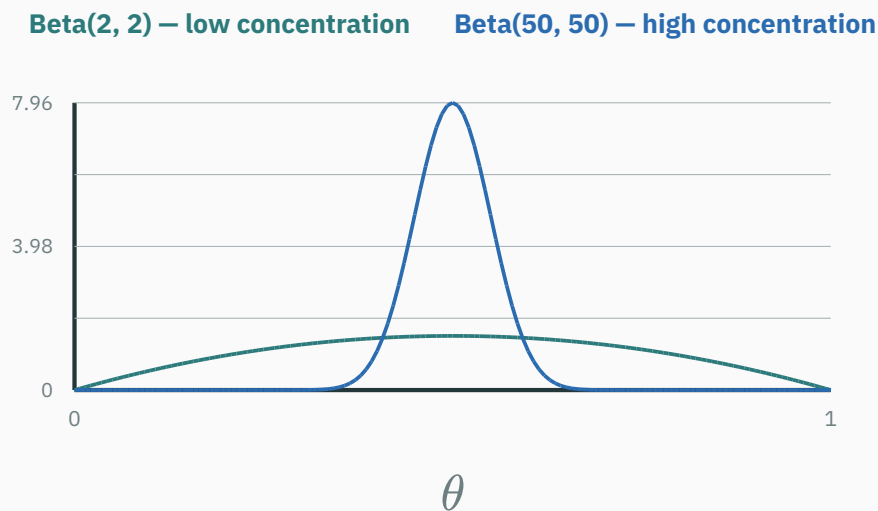
| Prior        | Imaginary record                      | Peak |
|--------------|---------------------------------------|------|
| Beta(1, 1)   | 0 H, 0 T — no memory at all           | flat |
| Beta(2, 2)   | 1 H, 1 T — the gentlest fairness hint | 0.5  |
| Beta(5, 2)   | 4 H, 1 T — suspects a heads bias      | 0.8  |
| Beta(50, 50) | 49 H, 49 T — fairness conviction      | 0.5  |

**For MAP,  $\alpha - 1$  and  $\beta - 1$  act like added head and tail counts.**

This is a mode mnemonic. The posterior update itself (next section) adds the raw counts  $n_H, n_T$  directly to  $\alpha$  and  $\beta$ .

# Concentration is controlled by $\alpha + \beta$

Same center, different confidence: mean =  $\alpha/(\alpha + \beta)$  fixes *where*; the total  $\alpha + \beta$  fixes *how firmly*.



A Beta(50, 50) prior is much more concentrated than Beta(2, 2), so the same dataset moves it less.

# Play the update before we derive it

**I** INTERACTIVE

INTERACTIVE

↗ [nipunbatra.github.io/interactive-articles/bayesian-posterior](https://nipunbatra.github.io/interactive-articles/bayesian-posterior)

Drag the prior's  $\alpha, \beta$ , stream in evidence across 8 experiments, and watch  $\text{prior} \times \text{likelihood} = \text{posterior}$  re-render live.

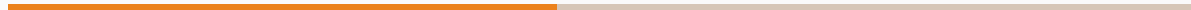
INTERACTIVE

↗ [seeing-theory.brown.edu/bayesian-inference/index.html](https://seeing-theory.brown.edu/bayesian-inference/index.html)

*Seeing Theory* · Bayesian Inference — flip a coin one toss at a time and watch a Beta posterior fill in.

Both toys are running one identity — the one we now derive by hand.

# Conjugacy: the one-line update



# Set up the multiplication

Data:  $n_H$  heads,  $n_T$  tails ( $n = n_H + n_T$ ). L14's likelihood, and a  $\text{Beta}(\alpha, \beta)$  prior — constants dropped, shapes kept ( $\propto$  absorbs  $1/B$ ):

$$p(\theta \mid D) \propto \underbrace{\theta^{n_H} (1 - \theta)^{n_T}}_{\text{likelihood}} \cdot \underbrace{\theta^{\alpha-1} (1 - \theta)^{\beta-1}}_{\text{prior density}}$$

Both factors speak the **same language**: powers of  $\theta$  and powers of  $(1 - \theta)$ . That coincidence is about to do all the work.

Same base, so multiplication is addition upstairs:

$$p(\theta \mid D) \propto \theta^{n_H + \alpha - 1} (1 - \theta)^{n_T + \beta - 1}$$

Stare at it: this is **the Beta shape again**, with new dials

$$\alpha' = \alpha + n_H, \quad \beta' = \beta + n_T$$

The head and tail counts add to the two exponents. The data do not change the family, only its parameters.



We hold a function  $\propto \theta^{\alpha'-1}(1-\theta)^{\beta'-1}$  that must integrate to 1 (it's a posterior). Among all rescalings  $c \cdot \theta^{\alpha'-1}(1-\theta)^{\beta'-1}$ , **exactly one** has area 1 — and  $\text{Beta}(\alpha', \beta')$  is that one, by definition.

So the constant is forced, no integral computed:

$$\text{Beta}(\alpha, \beta) \text{ prior} + n_H \text{ heads}, n_T \text{ tails} \Rightarrow \text{posterior} = \text{Beta}(\alpha + n_H, \beta + n_T)$$

“By inspection” means recognizing the Beta kernel and using its known normalizing constant.

# Conjugacy, named

**Definition.** A prior family is **conjugate** to a likelihood if the posterior always lands back in the same family.

- Beta in  $\rightarrow$  Bernoulli data  $\rightarrow$  Beta out: the Beta family is conjugate to coin flips.
- The entire update is **two integer additions**:  $\alpha \leftarrow \alpha + n_H$ ,  $\beta \leftarrow \beta + n_T$ . No integral, no grid, no optimizer.
- Our worked example: Beta(2, 2) + 3 heads  $\rightarrow$  Beta(5, 2), whose mode is 4/5.

# The MAP formula drops out

The posterior is  $\text{Beta}(\alpha + n_H, \beta + n_T)$ , and we know where a Beta peaks:

$$\theta_{\text{MAP}} = \frac{n_H + \alpha - 1}{n + \alpha + \beta - 2}$$

- Check: 3 heads,  $\text{Beta}(2, 2)$ :  $(3 + 1)/(3 + 2) = 4/5 = 0.8$  — matches the hand derivative. ✓
- Check: flat prior  $\alpha = \beta = 1$ :  $\theta_{\text{MAP}} = n_H/n = \theta_{\text{MLE}}$ . ✓

**The MAP estimate is the empirical count formula with the mode offsets  $\alpha - 1$  and  $\beta - 1$ .**

# Stress test: 9 heads, 1 tail

Ten flips, nine heads ( $\theta_{\text{MLE}} = 0.9$ ). Three believers meet the same data:

| Prior                     | $\theta_{\text{MAP}}$  | Reading                                       |
|---------------------------|------------------------|-----------------------------------------------|
| Beta(1, 1) — agnostic     | $9/10 = 0.900$         | no opinion → data speaks alone                |
| Beta(2, 2) — gentle hunch | $10/12 \approx 0.833$  | two pseudo-flips nudge back toward fair       |
| Beta(50, 50) — conviction | $58/108 \approx 0.537$ | 98 pseudo-flips: ten real ones barely dent it |

Same data, three defensible answers — the prior is a **published, criticizable assumption**, not a hidden thumb on the scale. (Which believer is being unreasonable? That argument is now about  $\alpha, \beta$  — in the open.)

# The conjugate catalog

The same algebra appears beyond coins:

| Likelihood                      | Conjugate prior | Posterior update                                               |
|---------------------------------|-----------------|----------------------------------------------------------------|
| Bernoulli (coin)                | Beta            | $\alpha \leftarrow \alpha + n_H, \beta \leftarrow \beta + n_T$ |
| Categorical ( $d$ faces)        | Dirichlet       | add each face's count to its pseudo-count                      |
| Gaussian mean ( $\sigma$ known) | Gaussian        | precision-weighted average — next section                      |

**Continuity ledger — planted today, paid off in L26:** the Dirichlet row is Beta's  $d$ -sided sibling. When L26's character language model meets an unseen letter pair, *Dirichlet smoothing* (add- $\alpha$  pseudo-counts) is exactly this table's second row.

**Prior Beta(2, 3); you then observe 4 heads and 1 tail. The posterior is...**

**A.** Beta(6, 4)

**B.** Beta(3, 7)

**C.** Beta(5, 2)

**D.** Beta(7, 8)

**Correct: A** — **Beta**(2 + 4, 3 + 1) = **Beta**(6, 4)

**Why:** Heads land on  $\alpha$ , tails on  $\beta$  — nothing else moves. B swapped the two counts; C threw the prior away and updated a flat Beta(1, 1); D added the total  $n = 5$  to both dials, double-counting every flip.

# Watching beliefs sharpen

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# Today's posterior is tomorrow's prior

Nothing forces the flips to arrive in one batch. Feed them one at a time, and **each posterior becomes the next prior**:

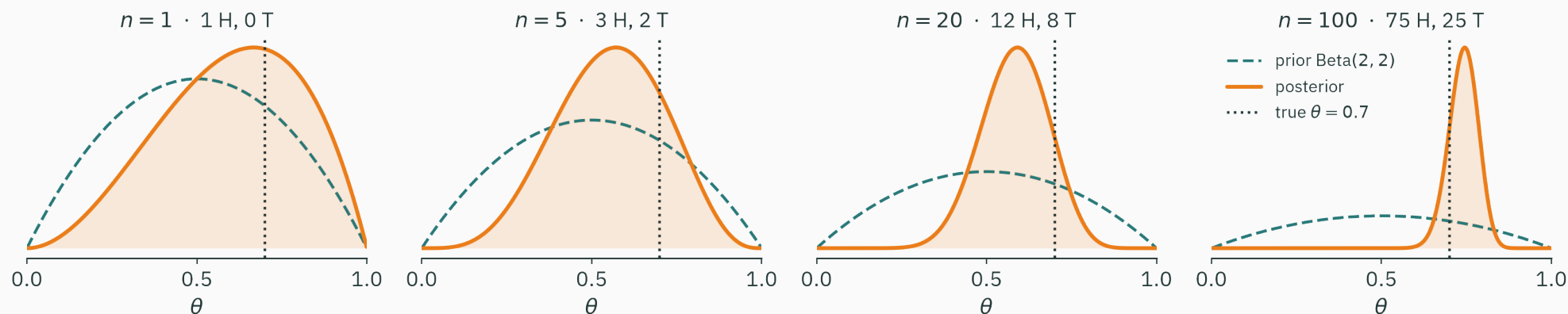
$$\text{Beta}(\alpha, \beta) \xrightarrow{\text{H}} \text{Beta}(\alpha + 1, \beta) \xrightarrow{\text{T}} \text{Beta}(\alpha + 1, \beta + 1) \xrightarrow{\text{H}} \dots$$

```
a, b = 2, 2          # Beta(2,2): 1 imaginary head, 1 tail
for y in flips:      # y = 1 for heads, 0 for tails
    a, b = a + y, b + 1 - y  # the ENTIRE Bayesian update
# at any moment: posterior = Beta(a, b), MAP = (a-1)/(a+b-2)
```

Two integers of memory, one addition per observation. This is **online learning** in its purest form — the model never re-reads old data.

# One hundred flips, four snapshots

A coin with true  $\theta = 0.7$  streams flips into a Beta(2, 2) prior:



After 1 flip the prior still dominates; by 100 flips (75 H, 25 T  $\rightarrow$  Beta(77, 27)) the posterior is a needle — parked at the **sample's** rate 0.75, a whisker above the truth. That residual wobble is the data's own sampling noise, not the prior's doing.

# The posterior sharpens like $1/\sqrt{n}$

How fast is certainty earned? The posterior's spread obeys

$$\text{sd} \approx \sqrt{\frac{\hat{\theta}(1 - \hat{\theta})}{n}}$$

- Quadruple the flips → **halve** the uncertainty — the same  $\sqrt{n}$  law you met in L13's CLT demo, now running *inside* the posterior.
- By  $n = 100$ , the prior's  $\alpha + \beta = 4$  pseudo-observations are 4 votes in an electorate of  $n + \alpha + \beta = 104$ : present, out-voted, and fading.

# Order cannot matter

Flip H then T, starting from  $\text{Beta}(\alpha, \beta)$ :  $(\alpha + 1, \beta)$ , then  $(\alpha + 1, \beta + 1)$ . Flip T then H:  $(\alpha, \beta + 1)$ , then  $(\alpha + 1, \beta + 1)$ . **Same place.**

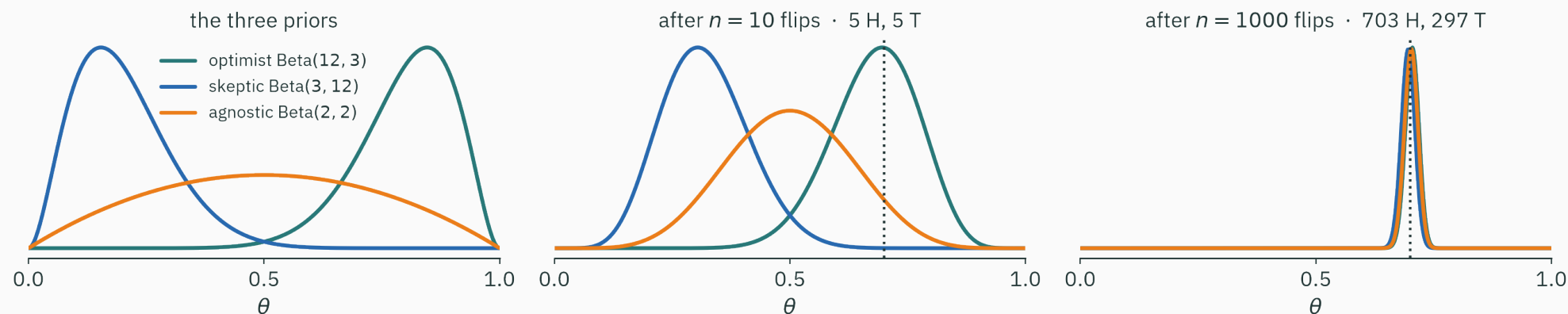
No accident: the posterior is a **product** of per-flip factors, and multiplication commutes —

$$p(\theta \mid D) \propto p(\theta) \prod_{i=1}^n P(y_i \mid \theta)$$

**Stream the data or dump it in one batch, in any order — **exactly** the same posterior.**

# Three scientists, three priors

An optimist, a skeptic, and an agnostic watch the same coin ( $\theta = 0.7$ ):



Ten flips in (5 H, 5 T), the posteriors still differ. After a thousand flips they are nearly indistinguishable.

# When the prior washes out

The formula says exactly when belief stops mattering:

$$\theta_{\text{MAP}} = \frac{n_H + \alpha - 1}{n + \alpha + \beta - 2} \rightarrow \frac{n_H}{n} = \theta_{\text{MLE}} \quad \text{as } n \rightarrow \infty$$

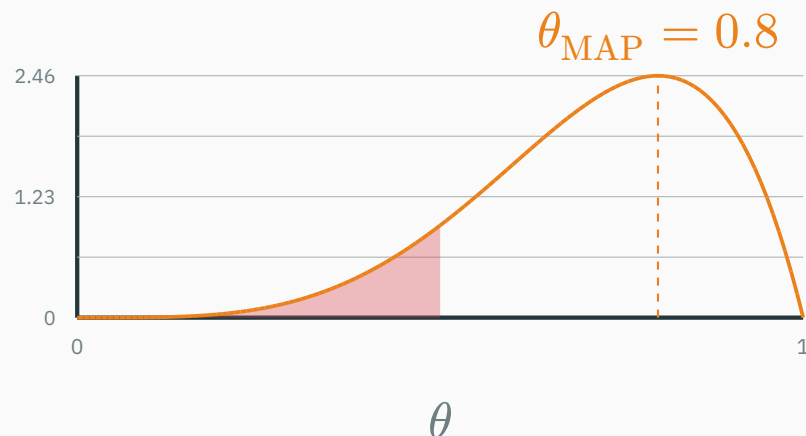
- The prior is a **fixed** number of pseudo-votes in a growing electorate: decisive at  $n = 3$ , a rounding error at  $n = 10^6$ .
- With a fixed regular prior and a growing i.i.d. sample, MLE and MAP typically converge to the same value.

**For this fixed Beta prior, the gap between MAP and MLE shrinks like  $1/n$ .**

The washout needs a **fixed** prior with **positive density near the truth**. A prior with zero support there, or pseudo-counts that grow with  $n$ , need not wash out.

# Posterior inference for the opening coin example

Three heads in three flips, Beta(2, 2) prior  $\rightarrow$  posterior Beta(5, 2):



- The MAP estimate is  $\theta_{\text{MAP}} = 0.8$ , while the posterior still assigns about 11% probability to  $\theta < 0.5$ .
- After three flips, the posterior remains uncertain and assigns nonzero probability to future tails.

## INTERACTIVE

↗ [nipunbatra.github.io/interactive-articles/mle-map-coin](https://nipunbatra.github.io/interactive-articles/mle-map-coin)

A coin you control: set the true bias, choose  $\alpha, \beta$ , and flip. Watch  $\theta_{\text{MAP}}$  glide from the prior's peak toward  $\theta_{\text{MLE}}$  as the pseudo-counts get out-voted.

Try a strong prior far from the data, then compare it with a flat prior at  $n = 3$ .



# Gaussian meets Gaussian

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# A continuous analogue

New estimation problem: a sensor reads a true value  $\mu$  through Gaussian noise (L13):

$$x_i \sim \mathcal{N}(\mu, \sigma^2), \quad \sigma \text{ known}, \quad i = 1, \dots, n$$

You also have a hunch — the calibration sheet says readings sit near  $\mu_0$ :

$$\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$$

Beta was the prior for a **probability**; the Gaussian is the prior for a **location**. And the Gaussian is conjugate to itself: Gaussian in  $\rightarrow$  Gaussian out.

## Stated: Gaussian in, Gaussian out

Multiply the two bells and complete the square (algebra in T7 and MML 8.3.2 — today, the shape of the answer). Call  $1/\text{variance}$  **precision** — confidence, as a number: prior precision  $\tau_0 = 1/\sigma_0^2$ , data precision  $\tau_D = n/\sigma^2$ .

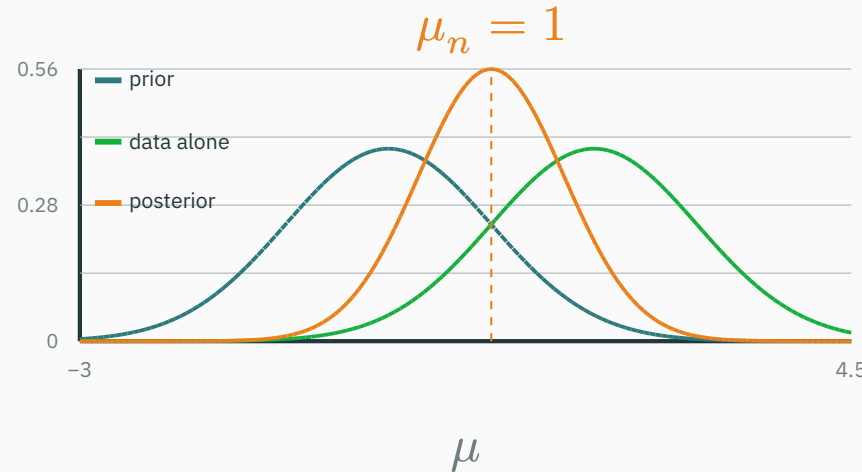
$$p(\mu \mid D) = \mathcal{N}(\mu_n, \sigma_n^2), \quad \frac{1}{\sigma_n^2} = \tau_0 + \tau_D$$

$$\mu_n = \frac{\tau_0 \mu_0 + \tau_D \bar{x}}{\tau_0 + \tau_D}$$

**Precisions add**, and the posterior mean is a **precision-weighted average** of hunch  $\mu_0$  and data average  $\bar{x}$  — the surer voice counts more.

Hunch  $\mu_0 = 0$  (precision  $1/\sigma_0^2 = 1$ ); data  $n = 4$ ,  $\bar{x} = 2$ ,  $\sigma^2 = 4$  (precision  $n/\sigma^2 = 1$ ).  
Equal confidence  $\rightarrow$  meet exactly halfway:

$$\mu_n = \frac{1 \cdot 0 + 1 \cdot 2}{1 + 1} = 1, \quad \sigma_n^2 = \frac{1}{1 + 1} = 0.5$$



The posterior is **narrower than both parents** — two independent confidences stacked.

# Precision is the currency

- Posterior mean = average of the two centers, **weighted by precision** — the surer voice counts more.
- The prior behaves like  $\sigma^2/\sigma_0^2$  **virtual observations** pinned at  $\mu_0$  — pseudo-counts again, in continuous clothing.
- As  $n$  grows, data precision  $n/\sigma^2 \rightarrow \infty$ , so a fixed finite prior precision has diminishing influence.

**Conjugate updating is an average where precision is the currency:  
pseudo-flips for coins, virtual readings for means.**

# MAP interpretations of common penalties

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## Recall the Gaussian-noise result from L14

Linear regression, Gaussian noise:  $y_i = \mathbf{w}^\top \mathbf{x}_i + \varepsilon_i$ ,  $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ . L14 showed:

$$\text{NLL}(\mathbf{w}) = \frac{1}{2\sigma^2} \|\mathbf{y} - X\mathbf{w}\|^2 + \text{const}$$

Under independent Gaussian observation noise with fixed variance, minimizing NLL is equivalent to least squares.

To form a MAP estimator, place an explicit prior on the weights.

# Give the weights a prior

Belief: weights shouldn't be huge — a model that needs  $w = 10^6$  is probably memorizing noise. Encode it:

$$\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma_0^2 I) \Rightarrow -\log p(\mathbf{w}) = \frac{1}{2\sigma_0^2} \|\mathbf{w}\|^2 + \text{const}$$

MAP swaps argmax-posterior for argmin of the negative log-posterior:

$$\mathbf{w}_{\text{MAP}} = \arg \min_{\mathbf{w}} \underbrace{\frac{1}{2\sigma^2} \|\mathbf{y} - X\mathbf{w}\|^2}_{\text{data misfit (L14)}} + \underbrace{\frac{1}{2\sigma_0^2} \|\mathbf{w}\|^2}_{\text{prior's fine on big weights}}$$

```
nll = ((y - X @ w)**2).sum() / (2 * sig2)    # L14: Gaussian NLL
nlp = (w**2).sum() / (2 * sig0_2)            # -log N(0, sig0^2) prior
loss = nll + nlp                             # MAP objective
```



# MAP with a Gaussian prior is ridge regression

Multiply through by  $2\sigma^2$  (argmins don't care):

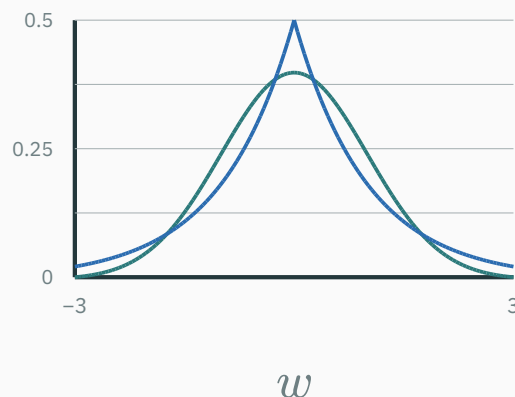
$$\mathbf{w}_{\text{MAP}} = \arg \min_{\mathbf{w}} \|\mathbf{y} - X\mathbf{w}\|^2 + \lambda \|\mathbf{w}\|^2, \quad \lambda = \frac{\sigma^2}{\sigma_0^2}$$

**Ridge regression is MAP with a Gaussian prior on the weights. The coefficient  $\lambda$  is the noise-to-prior-variance ratio.**

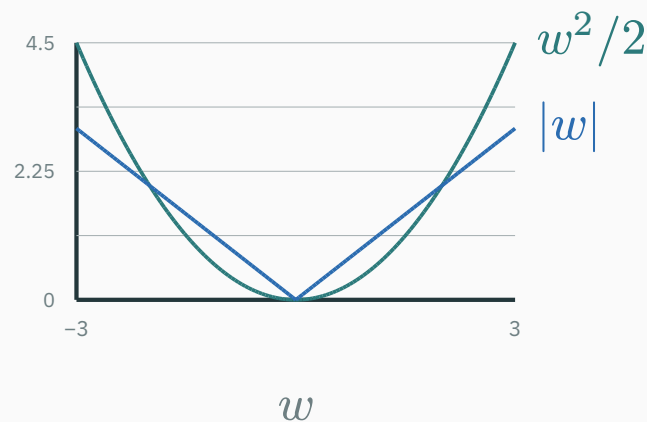
A later ML course may introduce ridge regression or weight decay as regularization, with  $\lambda$  tuned by validation. In this probabilistic model, increasing  $\lambda$  corresponds to decreasing the prior variance  $\sigma_0^2$  and concentrating more prior mass near zero.

A prior density contributes its negative logarithm to the MAP objective:

the prior — Gaussian vs Laplace



the penalty —  $\log p(w)$



Gaussian prior  $\rightarrow$  quadratic penalty  $\rightarrow$  **L2 / ridge**. Laplace prior  $\rightarrow |w|$  penalty  $\rightarrow$  **L1 / lasso**; the corner at zero allows exact zero-valued coefficients.

# The regularization dictionary

| You add to the loss...      | You just assumed...                         | Its street name                        |
|-----------------------------|---------------------------------------------|----------------------------------------|
| $\lambda \ w\ ^2$           | Gaussian prior $\mathcal{N}(0, \sigma_0^2)$ | ridge / weight decay                   |
| $\lambda \ w\ _1$           | Laplace prior                               | lasso                                  |
| add- $\alpha$ pseudo-counts | Beta / Dirichlet prior                      | Laplace smoothing ( $\rightarrow$ L26) |

If  $\exp(-R(w))$  is integrable, a penalty  $R$  can be interpreted as a negative log-prior in MAP estimation. Some training regularizers and constraints do not admit this simple interpretation.

# Summary

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# Lecture 15 — summary

- **Bayes**: posterior  $\propto$  likelihood  $\times$  prior — the evidence only normalizes.
- **MAP** climbs log-likelihood + log-prior; flat prior  $\Rightarrow$  MAP = MLE.
- **Beta**( $\alpha, \beta$ ): the MAP mode has offsets  $\alpha - 1, \beta - 1$ ;  $\alpha + \beta$  controls concentration.
- **Conjugacy** ★: Beta prior + coin data  $\rightarrow$  Beta( $\alpha + n_H, \beta + n_T$ ) — two additions.
- **Sequential = batch**, any order; width  $\sim 1/\sqrt{n}$ ; data swamps the prior.
- **Gaussians**: precisions add; posterior mean = precision-weighted average.
- **MAP penalty interpretations**: Gaussian  $\rightarrow$  ridge, Laplace  $\rightarrow$  lasso, Beta/Dirichlet  $\rightarrow$  smoothing.

**Read before L16** — MML 6.6.1 & 8.3.2 · MacKay Ch. 3. **T7** pairs with this lecture (grid-check every derivation); **Quiz 3 / midsem** covers Module 3.

# Practice problems

Try on paper; verify with the T7 notebook's grid trick.

1. Re-derive the ★ update:  $\text{Beta}(\alpha, \beta)$  prior,  $n_H$  heads,  $n_T$  tails  $\rightarrow$  posterior, from a blank page.
2. Prior  $\text{Beta}(5, 5)$ , data 2 H, 8 T: posterior,  $\theta_{\text{MAP}}$ , posterior mean — and compare with  $\theta_{\text{MLE}}$ .
3. Show that  $\text{Beta}(1, 1)$  makes  $\theta_{\text{MAP}} = \theta_{\text{MLE}}$  for **every** possible dataset.
4. How many **consecutive heads** push  $\theta_{\text{MAP}}$  above 0.99 under a  $\text{Beta}(2, 2)$  prior? (Solve  $(n + 1)/(n + 2) > 0.99$ .)
5. Sensor: prior  $\mathcal{N}(0, 1)$ ; data  $n = 9$ ,  $\bar{x} = 3$ ,  $\sigma^2 = 9$ . Posterior mean and variance?
6. A colleague doubles ridge's  $\lambda$ . What happens to  $\sigma_0^2$  in the MAP interpretation, and which prior would produce an L1 penalty instead?



## MAP keeps the peak, throws the width

★ OPTIONAL

MAP reports **one number** — but the posterior was a whole curve, and the curve knows more.

- Predicting the next flip by **averaging over the posterior** gives the posterior *mean*, not the mode:  $P(\text{next H} \mid D) = \frac{\alpha + n_H}{\alpha + \beta + n}$ .
- Three heads, Beta(2, 2): predict  $5/7 \approx 0.71$  — milder than  $\theta_{\text{MAP}} = 0.8$ , and a world away from MLE's 1.0.
- Keeping the **entire** posterior (and integrating over it) is Bayesian machine learning proper — the Probabilistic ML course lives there.



## Laplace's rule of succession

★ OPTIONAL

Flat prior + posterior mean  $\rightarrow P(\text{next H}) = \frac{n_H+1}{n+2}$  — “add one imaginary head and one imaginary tail.”

- Laplace, 1774: what is the probability the sun rises tomorrow, given  $n$  recorded sunrises? Answer:  $(n+1)/(n+2)$  — huge, but **never 1**.
- NLP's **add-one smoothing** is this rule, verbatim, applied to word counts — and its add- $\alpha$  generalization is the Dirichlet machinery L26 uses to keep the language model's  $\log 0$  at bay.

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250 years old, still shipping in production systems. Priors age well.



L2 and L1 penalties are negative log-priors in MAP estimation.

Next: **Module 4 — The Optimization Landscape**. We now know *what* to minimize; L16 begins the story of *how*.